MediCheck

Synthetic Test Document Data Specifications

Three Scenarios | SC + NC Region | BCBS SC / Ambetter / Molina | generate_test_data.py

PURPOSE & DATA COMPLIANCE

This document specifies the complete data fields for three synthetic medical bills and matching EOBs used as test and demonstration documents in MediCheck. All documents are generated programmatically by generate_test_data.py using the Python Faker library for patient data.

HIPAA compliance: No real patient data is used at any stage. All patient names, dates of birth, addresses, member IDs, and policy numbers are entirely fictitious. Provider names and NPI numbers are publicly available from the CMS NPPES registry. Insurance company names are publicly known. Each generated document includes a visible disclaimer: 'SYNTHETIC DOCUMENT — FOR DEMONSTRATION PURPOSES ONLY. All patient data is fictitious.'

CPT codes: CPT code numbers (not descriptions) are used to reference the CMS Physician Fee Schedule. CPT code descriptions are copyrighted by the AMA and are not reproduced in any document or system output.

MEDICARE RATE REFERENCE — CPT CODES USED

All Medicare rates below are approximate 2025 national facility rates from the CMS Physician Fee Schedule (PFS). South Carolina uses locality code 00 (statewide); Charlotte NC uses locality code 05. SC and NC rates are within 3-5% of national rates — use national rates for simplicity in Module 2 detection. When you download the CMS PFS CSV, verify and update these values; the important thing is that your injected error amounts reliably exceed the 300% threshold.

CPT Code	Service Description (paraphrased)	Approx. 2025 Medicare Rate (Facility)	Used In Scenario	Module Triggered
29881	Knee arthroscopy with meniscectomy	~$865	A (surgeon bill)	Module 2 — rate outlier check
00400	Anaesthesia — integumentary procedures	~$285 (base + time units)	A (anaesthesia bill)	Module 4 — NSA violation; Module 2 — upcoding
27447	Total knee arthroplasty (TKA)	~$1,450	A (facility — for context only)	—
72193	CT pelvis with contrast	~$145	B (radiology bill)	Module 3 — EOB mismatch
72191	CT angiography, pelvis	~$285	B (EOB — different code)	Module 3 — EOB mismatch
29881 (dup)	Knee arthroscopy — duplicate line	~$865	B (MUSC facility bill)	Module 1 — duplicate charge

99205	New patient office visit, high complexity	~$210	C (physician bill — upcoded)	Module 2 — rate outlier
99213	Established patient office visit, low-med complexity	~$87	C (correct code — for reference)	Module 2 — correct rate baseline
A4570	Surgical supply — sterile tray	~$45	A (facility — duplicate)	Module 1 — duplicate charge

How to use these rates: For Module 2 to fire, your billed amount must exceed 300% of the Medicare rate. Example: CPT 29881 Medicare rate ~$865. To trigger Module 2, set billed amount to $3,200+ (370% of $865). The exact multiplier doesn't matter as long as it clears 300%. Use round numbers that look like real hospital bills.

Scenario A — Crossborder Surgery — Rock Hill SC Patient at Charlotte NC Hospital

All four error detection modules fire in this scenario. It is the primary demo document — use it first in the presentation.

Errors injected: 1. Duplicate surgical supply charge (Module 1). 2. Anaesthesia time upcoded — 180 minutes billed, 90 minutes documented (Module 2). 3. Surgeon fee exceeds 300% Medicare rate (Module 2). 4. Anaesthesiologist is out-of-network — No Surprises Act violation (Module 4). 5. EOB total does not match sum of line items on bill (Module 3).

A1 — Patient Data (Faker-generated)

Field	Synthetic Value	Faker Method
Patient name	James R. Whitfield	fake.name() — seed 42
Date of birth	March 14, 1979 (age 45)	fake.date_of_birth(minimum_age=44, maximum_age=46)
Address	218 Cherry Road, Rock Hill, SC 29732	fake.street_address() + 'Rock Hill, SC 29732'
Phone	(803) 555-0142	fake.numerify('(803) 555-0###')
Email	j.whitfield.rh@fakemail.test	fake.email() — domain forced to .test
Member ID	BCB-SC-4419-7823	fake.numerify('BCB-SC-####-####')
Policy number	GRP-774-SC-2024	fake.numerify('GRP-###-SC-2024')
Group name	Whitfield Industrial Services, Inc.	Hardcoded fictional employer
SSN (partial, last 4 only)	XXXXX-7341	fake.numerify('XXXXX-####')

A2 — Insurance / Payer Data

Field	Value
Insurer	BlueCross BlueShield of South Carolina
Plan name	BlueChoice PPO Gold — Employer Group
Payer ID	38000
Claims address	P.O. Box 100190, Columbia, SC 29202
Customer service phone	1-800-868-2522
Plan type	PPO — in-network and out-of-network benefits apply
Deductible (YTD met)	$1,500 individual — $1,200 met as of date of service
Out-of-pocket max	$4,000 individual — $1,200 met
Coinsurance (in-network)	20% after deductible
Out-of-network coinsurance	40% after out-of-network deductible — BEFORE NSA application

A3 — Provider Data (NPPES Public Registry)

Provider	Entity Name	Address	NPI (Public)	Network Status
Hospital (facility)	Atrium Health Carolinas Medical Center	1000 Blythe Blvd, Charlotte, NC 28203	1174675214	IN-NETWORK with BCBS SC
Surgeon	Dr. Michael T. Forsythe, MD (fictional attending — use Atrium group NPI)	1000 Blythe Blvd, Charlotte, NC 28203	Use facility NPI	IN-NETWORK with BCBS SC
Anaesthesiologist	Carolina Anesthesia Associates, PA (synthetic entity)	2001 Vail Ave, Charlotte, NC 28207 (synthetic address)	Use synthetic NPI: 9999000001	OUT-OF-NETWORK — triggers NSA violation
Referring physician	Dr. Patricia L. Odom, MD (synthetic — Rock Hill SC PCP)	412 Oakland Ave, Rock Hill, SC 29730 (synthetic)	Use synthetic NPI: 9999000002	IN-NETWORK

Synthetic NPIs: Prefix all synthetic (non-real) NPIs with 9999 to clearly distinguish them from real public NPIs in your code. A simple validation check in generate_test_data.py can flag any NPI starting with 9999 as synthetic.

A4 — Date of Service & Encounter Details

Field	Value
Date of service	September 17, 2025
Place of service code	21 (Inpatient Hospital)
Admission date	September 17, 2025
Discharge date	September 18, 2025 (1-night stay)
Primary diagnosis code (ICD-10)	M23.61 — Other spontaneous disruption of anterior cruciate ligament, right knee
Secondary diagnosis	M25.361 — Stiffness of right knee, not elsewhere classified
DRG (facility billing)	DRG 483 — Major joint replacement or reattachment of lower extremity w/o MCC
Surgical procedure performed	Right knee arthroscopy with partial medial meniscectomy
Anaesthesia type	General — endotracheal
Documented anaesthesia time	90 minutes (per anaesthesia record — see injection note)

A5 — Provider Bill Line Items (Scenario A — 3 Separate Bills)

Bill A-1: Atrium Health Carolinas Medical Center — Facility Bill

Line	Service	CPT	Qty	Unit $	Total Billed
1	Arthroscopy knee — right (surgeon fee)	29881	1	$3,200.00	$3,200.00
2	OR facility fee	00790	1	$4,800.00	$4,800.00
3 ⚠	Surgical supply — sterile tray (FIRST CHARGE)	A4570	1	$480.00	$480.00
4	Inpatient room & board (1 night)	99238	1	$2,200.00	$2,200.00
5	Radiology — knee X-ray	73560	2	$320.00	$640.00
6	Physical therapy evaluation (post-op)	97001	1	$285.00	$285.00
7 ⚠	Surgical supply — sterile tray (DUPLICATE — same item as Line 3)	A4570	1	$480.00	$480.00
	TOTAL BILLED				**$12,085.00**

Bill A-2: Carolina Anesthesia Associates, PA — Anaesthesia Bill (separate provider, out-of-network)

Line	Service	CPT	Units	Unit $	Total Billed
1 ⚠	Anaesthesia — lower extremity surgery. BILLED: 18 units (180 min). DOCUMENTED: 9 units (90 min).	00400	18	$95.00	$1,710.00 ⚠ UPCODED
2	Base anaesthesia units (standard 7 base units)	00400-BP	7	$95.00	$665.00
	TOTAL BILLED				**$2,375.00**

Module 4 trigger: Bill A-2 must include a network status field showing 'Out-of-Network Provider' in the field confirmation UI. The EOB will show the NSA flag. Carolina Anesthesia Associates' NPI (9999000001) must appear in your synthetic out-of-network provider list in the knowledge base ingestion.

A6 — EOB Data (BCBS SC — Scenario A)

The EOB is issued by BlueCross BlueShield of South Carolina after processing both bills. It contains a deliberate mismatch on the sterile tray line to trigger Module 3.

EOB Field	Value
EOB document date	October 3, 2025
Claim number	CLM-38000-2025-094412
Patient name	James R. Whitfield
Member ID	BCB-SC-4419-7823
Provider (primary)	Atrium Health Carolinas Medical Center
Provider (anaesthesia)	Carolina Anesthesia Associates, PA — OUT-OF-NETWORK
Total amount billed (per EOB)	$13,960.00

Insurer negotiated rate (in-network)	$7,240.00
Plan paid (in-network portion)	$4,832.00
Member responsibility (in-network)	$2,408.00
NSA flag	BALANCE BILLING PROHIBITED — No Surprises Act applies to anaesthesia services at in-network facility. Member not liable for out-of-network differential.
EOB sterile tray line (Module 3 mismatch)	EOB shows A4570 billed ONCE at $480.00 — provider bill shows TWICE at $480.00 each. EOB total does not match provider bill total by $480.00.

Scenario B — Outpatient Arthroscopy — Charleston SC Patient at MUSC Health

Modules 1, 2, and 3 fire in this scenario. Best for demonstrating the EOB reconciliation module with a clear radiology code mismatch.

Errors injected: 1. Duplicate knee arthroscopy charge on facility bill — same CPT 29881 on same date (Module 1). 2. MUSC facility fee at ~370% of Medicare rate (Module 2). 3. Radiology CPT code on provider bill (72193) differs from EOB (72191) — wrong procedure code billed (Module 3).

B1 — Patient Data (Faker-generated)

Field	Synthetic Value	Faker Method
Patient name	Rebecca A. Tanner	fake.name() — seed 84
Date of birth	July 22, 1986 (age 38)	fake.date_of_birth(minimum_age=37, maximum_age=39)
Address	1847 Folly Road, Charleston, SC 29412	fake.street_address() + 'Charleston, SC 29412'
Phone	(843) 555-0267	fake.numerify('(843) 555-0###')
Member ID	AMB-SC-7731-0054	fake.numerify('AMB-SC-####-####')
Policy number	ATC-SILVER-2025-3341	fake.numerify('ATC-SILVER-2025-####')
Group / plan	Ambetter Silver HMO — Individual Marketplace Plan	Hardcoded

B2 — Insurance / Payer Data

Field	Value
Insurer	Ambetter from Absolute Total Care
Payer ID	68069
Claims address	P.O. Box 21100, Eagan, MN 55121
Plan type	HMO — requires PCP referral; in-network only (no out-of-network benefit except emergencies)
Deductible (annual)	$3,212 individual — $0 met at time of service (first claim of year)
Coinsurance	30% after deductible
Out-of-pocket max	$5,545

B3 — Provider Data

Provider	Entity Name	Address	NPI	Network Status

Hospital (facility)	MUSC Health University Medical Center	169 Ashley Ave, Charleston, SC 29425	1437158858	IN-NETWORK with Ambetter
Surgeon	Dr. James L. Beaumont, MD (synthetic attending)	96 Jonathan Lucas St, Charleston, SC 29425	Synthetic: 9999000003	IN-NETWORK with Ambetter
Radiology group	Lowcountry Radiology Consultants, LLC (synthetic)	135 Rutledge Ave, Charleston, SC 29403 (synthetic)	Synthetic: 9999000004	IN-NETWORK with Ambetter

B4 — Date of Service & Encounter Details

Field	Value
Date of service	November 6, 2025
Place of service code	22 (Outpatient Hospital)
Primary diagnosis (ICD-10)	M23.202 — Derangement of unspecified meniscus due to old tear or injury, left knee
Procedure performed	Left knee arthroscopy with partial lateral meniscectomy
Imaging ordered	CT arthrogram of left knee (ordered same day, performed same day)

B5 — Provider Bill Line Items

Bill B-1: MUSC Health University Medical Center — Facility + Surgeon Bill

Line	Service	CPT	Qty	Unit $	Total Billed
1	Arthroscopy left knee — surgical fee	29881	1	$3,200.00	$3,200.00
2	Outpatient facility fee — operating room	00820	1	$5,600.00	$5,600.00
3	Recovery room	99217	1	$680.00	$680.00
4 ⚠	Arthroscopy left knee — DUPLICATE (same procedure, same date as Line 1)	29881	1	$3,200.00	$3,200.00 ⚠ DUPLICATE
	TOTAL BILLED				**$12,680.00**

Bill B-2: Lowcountry Radiology Consultants, LLC — Radiology Bill (separate provider)

Line	Service	CPT	Qty	Unit $	Total Billed
1 ⚠	CT pelvis with contrast (WRONG CODE — see EOB mismatch)	72193	1	$1,840.00	$1,840.00 ⚠ CODE MISMATCH
	TOTAL BILLED				**$1,840.00**

B6 — EOB Data (Ambetter — Scenario B)

EOB Field	Value
EOB document date	November 24, 2025
Claim number	CLM-68069-2025-118843
Patient name	Rebecca A. Tanner
Member ID	AMB-SC-7731-0054
Total amount billed (per EOB)	$9,480.00
Note on duplicate	EOB only shows one 29881 charge — duplicate was caught by insurer and denied. Denial reason: CO-4 (service not payable per contract). This is intentional — Module 1 still fires because the provider bill has both lines.
Radiology code on EOB (Module 3 mismatch)	EOB shows CPT 72191 (CT angiography, pelvis — Medicare rate ~$285) was processed. Provider bill shows CPT 72193 (CT pelvis with contrast — Medicare rate ~$145). Different procedures. Code mismatch triggers Module 3.
MUSC facility fee on EOB	$3,200.00 allowed (vs $5,600.00 billed — 175% of Medicare rate — does NOT trigger Module 2 at 300% threshold)
Surgeon fee on EOB	$865.00 allowed (vs $3,200.00 billed — 370% of Medicare rate — TRIGGERS Module 2)

Scenario C — Primary Care Upcoding — Greenville SC Patient at Prisma Health Medical Group

Module 2 fires in this scenario (rate outlier indicating potential upcoding). This is the simplest scenario — single bill, single EOB, one clear error. Best for demonstrating Module 2 in isolation or as the third document in a sequence.

Error injected: Visit billed as CPT 99205 (new patient, high complexity, Medicare rate ~$210). Patient is an established patient whose record shows routine diabetes follow-up — correct code is CPT 99213 (established patient, low-medium complexity, Medicare rate ~$87). Billed amount of $340 represents 161% of the correct Medicare rate but 322% of what should have been billed — triggers Module 2 outlier flag against the correct baseline code.

Why this triggers Module 2: Your Module 2 compares the billed CPT code's amount against the Medicare rate for that same CPT code. CPT 99205 Medicare rate is ~$210. Bill shows $340 for 99205. $340 / $210 = 162% — this does NOT trigger the 300% threshold on its own. To make this scenario reliably trigger Module 2, set the billed amount to $680 (324% of $210). The RAG explanation then notes this is consistent with upcoding from 99213 ($87) to 99205 ($210), and the $680 charge is also 782% of the 99213 Medicare rate — a clear anomaly.

C1 — Patient Data (Faker-generated)

Field	Synthetic Value	Faker Method
Patient name	Dorothy M. Simmons	fake.name() — seed 117
Date of birth	February 3, 1972 (age 52)	fake.date_of_birth(minimum_age=51, maximum_age=53)
Address	3104 Augusta Road, Greenville, SC 29605	fake.street_address() + 'Greenville, SC 29605'
Phone	(864) 555-0391	fake.numerify('(864) 555-0###')
Member ID	MOL-SC-2241-8819	fake.numerify('MOL-SC-####-####')
Policy number	MHC-GOLD-2025-7714	fake.numerify('MHC-GOLD-2025-####')
Established patient since	March 2019 — 6-year established patient (critical for upcoding scenario)	Hardcoded

C2 — Insurance / Payer Data

Field	Value
Insurer	Molina Healthcare of South Carolina
Payer ID	38217
Claims address	P.O. Box 22818, Long Beach, CA 90801
Plan type	HMO — Medicaid-adjacent marketplace plan; in-network only
Deductible	$0 (Gold plan — no deductible)

Copay for primary care visit	$30 per visit
Out-of-pocket max	$2,500

C3 — Provider Data

Provider	Entity Name	Address	NPI	Network Status
Physician group (facility)	Prisma Health Medical Group	300 E McBee Ave, Greenville, SC 29601	1790786653 (system NPI)	IN-NETWORK with Molina Healthcare
Attending physician	Dr. Sarah M. Harrington, MD (synthetic)	300 E McBee Ave, Greenville, SC 29601 (same address)	Synthetic: 9999000005	IN-NETWORK with Molina Healthcare

C4 — Date of Service & Encounter Details

Field	Value
Date of service	October 21, 2025
Place of service code	11 (Office)
Primary diagnosis (ICD-10)	E11.9 — Type 2 diabetes mellitus without complications
Visit type (per encounter notes)	Established patient routine follow-up — diabetes management review, medication check, HbA1c result discussion. Duration: 15 minutes. Medical decision making: straightforward.
Correct CPT code	99213 — Established patient, office visit, low to medium complexity (Medicare rate ~$87)
Billed CPT code (error)	99205 — New patient, high complexity (Medicare rate ~$210). Patient is NOT new — established since 2019.

C5 — Provider Bill Line Items

Line	Service	CPT	Qty	Unit $	Total Billed
1 ⚠	Office visit — new patient, high complexity (UPCODED: established patient, routine follow-up)	99205	1	$680.00	$680.00 ⚠ RATE OUTLIER
2	HbA1c blood test	83036	1	$42.00	$42.00
3	Blood glucose quantitative	82947	1	$28.00	$28.00
	TOTAL BILLED				**$750.00**

C6 — EOB Data (Molina Healthcare — Scenario C)

EOB Field	Value

EOB document date	November 5, 2025
Claim number	CLM-38217-2025-108871
Patient name	Dorothy M. Simmons
Member ID	MOL-SC-2241-8819
Total billed (per EOB)	$750.00
Molina allowed amount	$520.00 (in-network contracted rate for 99205)
Plan paid	$490.00 ($520 allowed minus $30 copay)
Member responsibility	$30.00 (copay only — no deductible on Gold plan)
Note	EOB and provider bill match on CPT code 99205 — the upcoding is not caught by the insurer. Module 2 catches it by comparing the billed amount against the Medicare rate for 99205, which reveals a 324% markup over what Medicare would pay, and the RAG explanation notes the encounter diagnosis and duration are inconsistent with high-complexity new patient billing.

GENERATE_TEST_DATA.PY — IMPLEMENTATION SPECIFICATION

The following is the complete specification for the Python script that generates all three synthetic document sets as PDFs. Place the script in /test-data/generate_test_data.py in the repo.

Dependencies

```
pip install faker reportlab python-barcode
```

Script Structure

- Set Faker seed at top of file: fake = Faker(); Faker.seed(42). This ensures documents are reproducible — the same seed always generates the same names and addresses.
- Define three ScenarioConfig dataclasses: ScenarioA, ScenarioB, ScenarioC. Each contains all fields from this spec as typed attributes.
- Define a BillRenderer class using ReportLab's canvas.Canvas. Methods: render_header(), render_patient_block(), render_line_items_table(), render_totals(), render_disclaimer().
- Define an EOBRenderer class similarly. EOBs have a different visual layout — two-column summary block at top, line items below, NSA/denial notices at bottom.
- Main function generates 6 PDFs total: scenario_a_bill.pdf, scenario_a_eob.pdf, scenario_b_bill.pdf, scenario_b_eob.pdf, scenario_c_bill.pdf, scenario_c_eob.pdf.
- All PDFs saved to /test-data/synthetic/ directory. This directory is in .gitignore if you prefer not to commit PDFs, or committed if you want them always available.

Disclaimer Block — Required on Every Document

Required text: Every generated PDF must include a visible box at the bottom of page 1 with the following text: 'SYNTHETIC DOCUMENT — FOR DEMONSTRATION PURPOSES ONLY. All patient data (names, dates of birth, addresses, member IDs) is entirely fictitious and generated using the Python Faker library. Provider names and NPI numbers are from the publicly available CMS NPPES registry. Insurance company names are publicly known. This document does not represent a real patient record and was not derived from any real medical encounter. Generated by MediCheck generate_test_data.py.'

Error Injection Verification — assert() Checks

Add these assertions at the end of generate_test_data.py to verify errors are correctly injected before the PDFs are committed:

```
# Scenario A — verify duplicate A4570
assert scenario_a.bill_lines.count(cpt='A4570') == 2, 'Scenario A: duplicate not
injected'

# Scenario A — verify anaesthesia upcoding
assert scenario_a.anesthesia_units_billed == 18, 'Scenario A: units should be 18
(billed)'
assert scenario_a.anesthesia_units_documented == 9, 'Scenario A: units should be 9
(documented)'

# Scenario B — verify duplicate 29881
assert scenario_b.bill_lines.count(cpt='29881') == 2, 'Scenario B: duplicate 29881 not
injected'
```

```
# Scenario B — verify radiology code mismatch
assert scenario_b.radiology_bill_cpt == '72193', 'Scenario B: bill CPT should be
72193'
assert scenario_b.radiology_eob_cpt  == '72191', 'Scenario B: EOB CPT should be 72191'

# Scenario C — verify upcoding amount triggers 300% threshold
assert scenario_c.billed_amount_99205 / 210 > 3.0, 'Scenario C: must exceed 300% of
Medicare rate'
print('All error injection assertions passed.')
```

Running the Script

- From repo root: python test-data/generate_test_data.py
- Output: test-data/synthetic/*.pdf (6 files total)
- Re-run any time to regenerate clean documents
- Include in CI/CD pipeline as an optional step: generate docs, then run integration tests against them

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These three scenarios are your demo documents, your integration test fixtures, and your Module validation suite — all from the same reproducible script. Build them once, test them constantly.